Rafael Allan D. Alvarado

CS-203

Module 4 – Process Modelling Assignment:

1. Explain the four elements given in the figure below. Discuss also each notation Provided.



Four Elements of Process Modelling

External entity
An external entity is a person, institution, or department outside of the modeled system. It is a system in which data originates (data source) and usually ends (data sink).

Process
The process is a system's input-to-output transformation that results from a process function. Process symbols might be rectangular with rounded corners, oval, rectangle, or circle. The process is referred to as a brief sentence, in one word or phrase to explain its essence.

Data store
A data store is a collection of data saved in a warehouse for future use. The store symbol is made up of two horizontal lines. The warehouse is not limited to being a data file. It can be anything from a folder of documents to an optical disc to a filing cabinet. The data warehouse can be seen independently of how it was implemented.

Data flow
Data flow outlines how information is transferred between different elements of a system. The arrow is a symbol for data transmission. To identify the information being transported, give the flow a relatable name.

Four notations of process Modelling

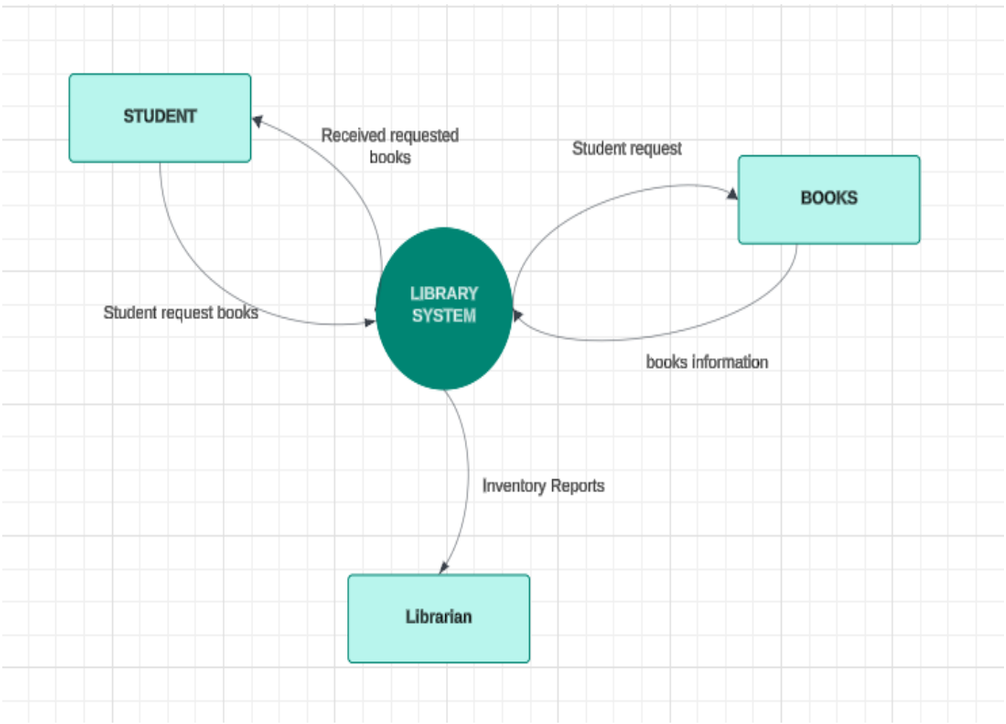
Yourdon and DeMarco notation
The Yourdon and DeMarco notation is a graphical depiction of a system's data flow, depicting how data moves between processes, external entities, and data stores. It is utilized for structured system analysis and understanding a system's functional breakdown.

Gane & Sarson notation
It is used to display system operations, external entities, data stores, and data flows, assisting analysts and designers in understanding system functionality. It differs from Yourdon and DeMarco's notation by the forms used to express processes and things.

SSADM Notation
It is a structured method for information system analysis, design, and documentation. It provides an organized framework for employing standardized diagrams to depict system interactions, data flows, and processes.

Unified Notation
It's a standardized visual notation for representing software and business processes. UML offers a uniform method for modeling system structures, behaviors, and interactions in a consistent and orderly manner.

2. Among the four(4) notation given, choose one and make your own example on how does this notation is applied in presenting the flow of your data. Apply the chosen notation in creating your first diagram in DFD (Context Diagram).



In a level 0 data flow diagram using the **Yourdon and DeMarco notation**, the process begins when the student requests books from the library system. The library system retrieves the requested book's information from its database and then gives it to the students. Then, the student receives the requested book. Finally, the librarian receives an inventory report for the transaction.